

Taxation and the Welfare State: Solutions to Problem Set 10

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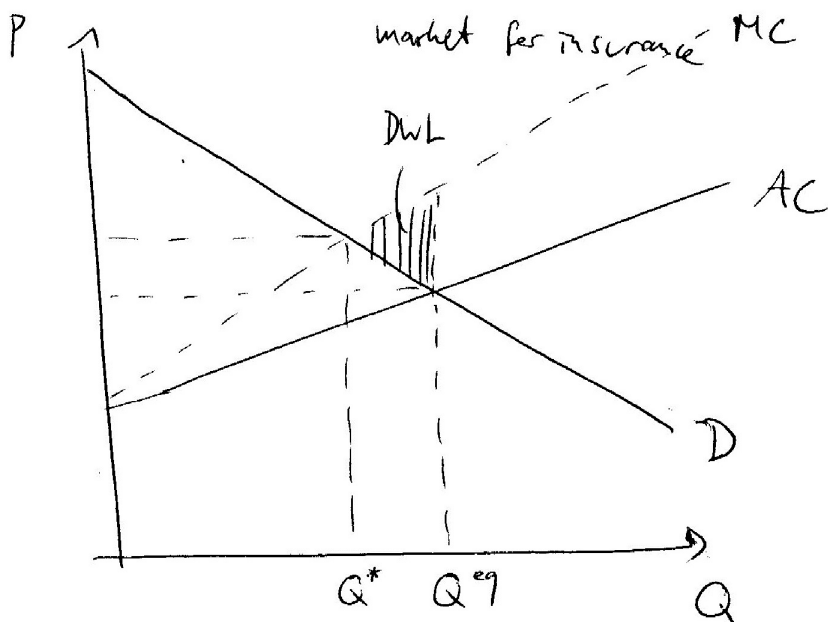
Question 1

a.

According to the results of Finkelstein & McGarry (2006), individuals with lower risk have a higher willingness to pay. This is a situation of advantageous selection.

Under what condition can advantageous selection arise? Recall from the lecture that we can write an individual's valuation of insurance as $v_i = \pi_i + r_i$, where π_i reflects the individual's risk and r_i reflects preferences for insurance (other than risk). Advantageous selection can only arise there is a strongly negative correlation between π_i and r_i , such that lower-risk individuals have a higher overall valuation of insurance.

b.



With advantageous selection, cost curves are upward-sloping, and MC is above AC . The efficient allocation would require $D = MC$, but the competitive equilibrium is where $D = AC$. There is over-insurance, i.e. too many individuals buy insurance, and the price is below the efficient price. The resulting deadweight loss equals the shaded triangle.

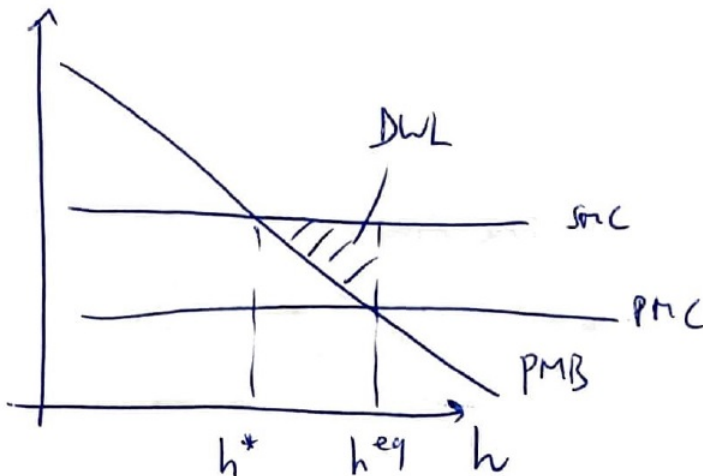
c.

A subsidy would exacerbate the inefficiency. By shifting the demand curve upwards, it would lead to an even higher quantity.

As with adverse selection, policy interventions could take the form of both quantity and price regulation. An example of quantity regulation could be a ban on insurance (the analogue of a mandate under adverse selection). However, this would entail a welfare loss equal to the triangle between D and MC from 0 to Q^* . Price regulation in terms of a Pigouvian tax can achieve the optimal quantity (shift D curve downwards by tax). In addition, a Pigouvian tax would generate revenue for the government.

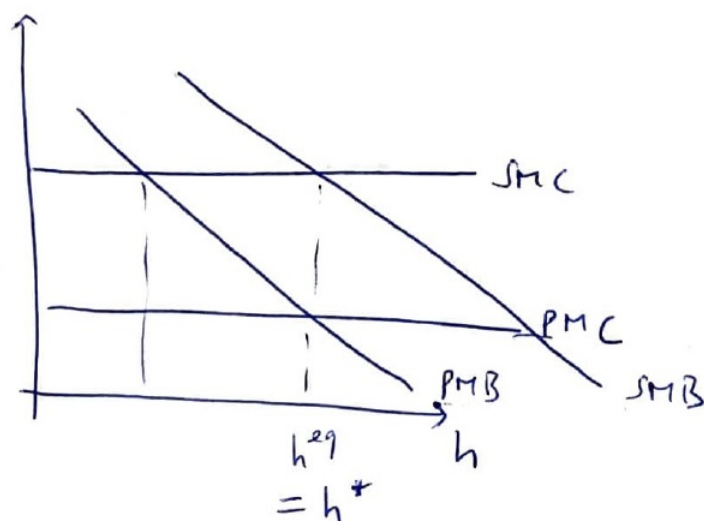
Question 2

a.



In the diagram, the private marginal cost is below the social marginal cost due to health insurance that features some cost sharing. Consumers who have insurance do not have to pay the true marginal cost of healthcare, but only a smaller price as specified in their insurance scheme. This price could even be zero with full insurance. Thus, there is overconsumption of healthcare, $h^{eq} > h^*$, and a deadweight loss marked by the shaded triangle in the figure arises.

b.



Positive externalities from healthcare consumption could arise due to a number of reasons, including in the case of infectious diseases, where treating or vaccinating one individual protects other individuals. Better health could also lead to higher productivity and fiscal externalities.

In a situation with positive externalities, the social marginal benefit from healthcare is higher than the private marginal benefit. As shown in the graph, this would lead to underconsumption of healthcare when individuals face the full marginal cost of healthcare. In this case, insurance can correct the inefficiency due to externalities. When individuals face a lower price of healthcare due to insurance, they increase healthcare consumption towards the social optimum.

c.

An internality in healthcare consumption can be due to myopia or lack of information, for instance, such that consumers do not fully take into account the future benefits of medical treatment. In principle, this can be modeled like a positive externality. Besides the part of the benefit from healthcare internalized by the consumer, there is an additional positive internality on the same consumer. An internality can be corrected for with a subsidy via (social) insurance just like an externality. Normatively, however, correcting internalities is more controversial than correcting externalities. The “paternalism critique” applies: Does the government really know better than people themselves what is good for them?

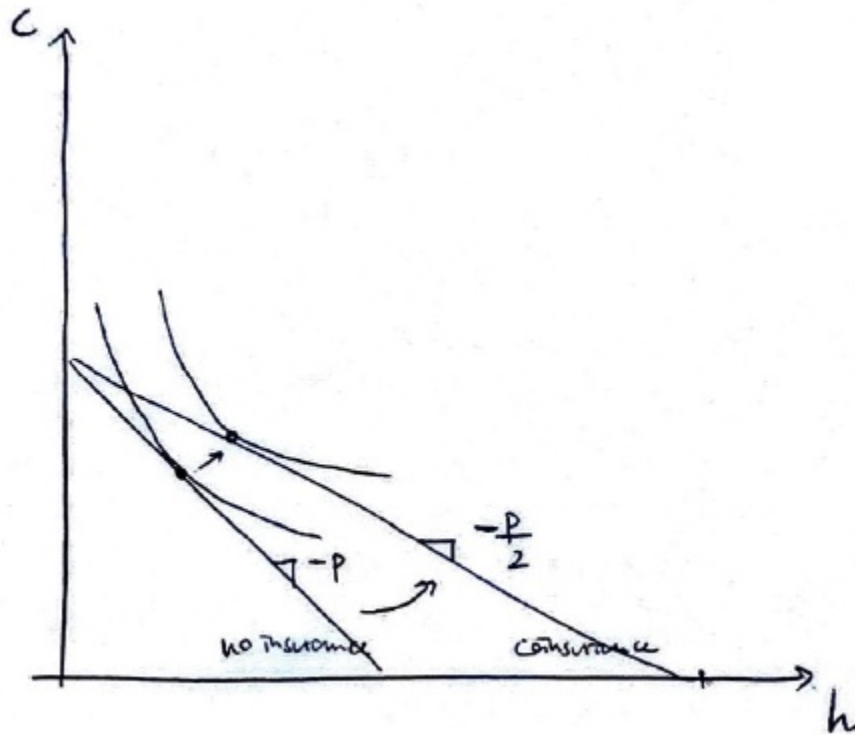
Question 3

a.

- Deductible: The patient has to pay the first €X of healthcare expenditure
- Copayment: The patient has to pay €X for each unit of healthcare

- Coinsurance: The patient has to pay $X\%$ of healthcare expenditure

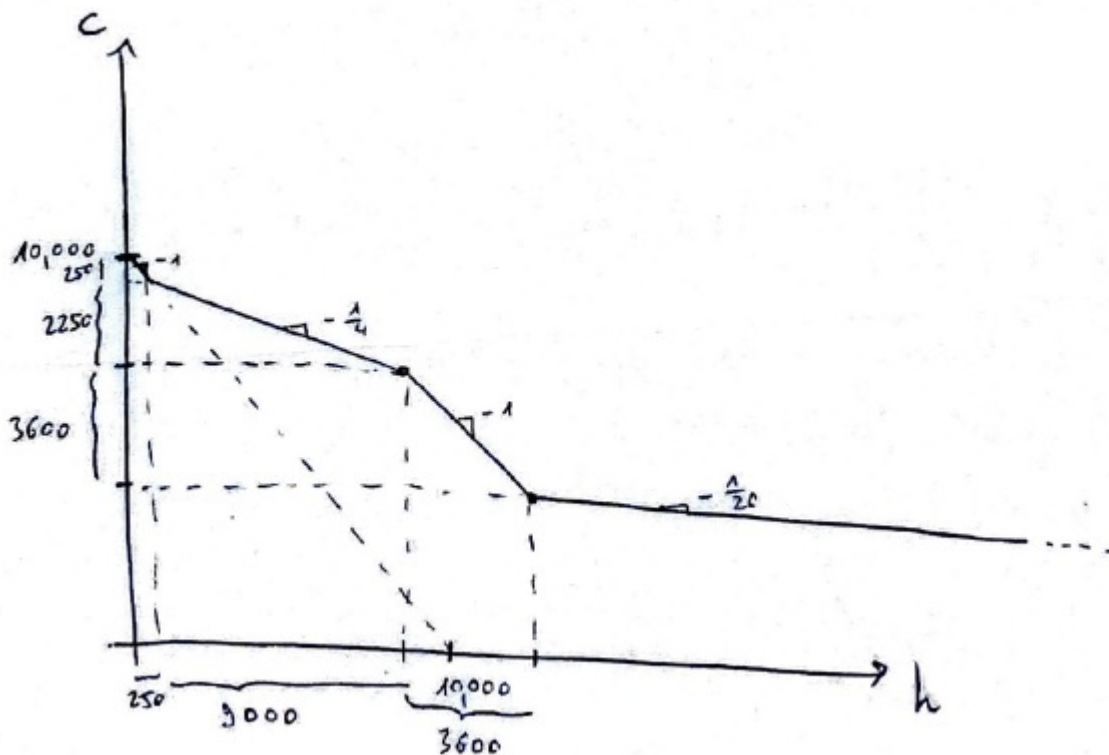
b.



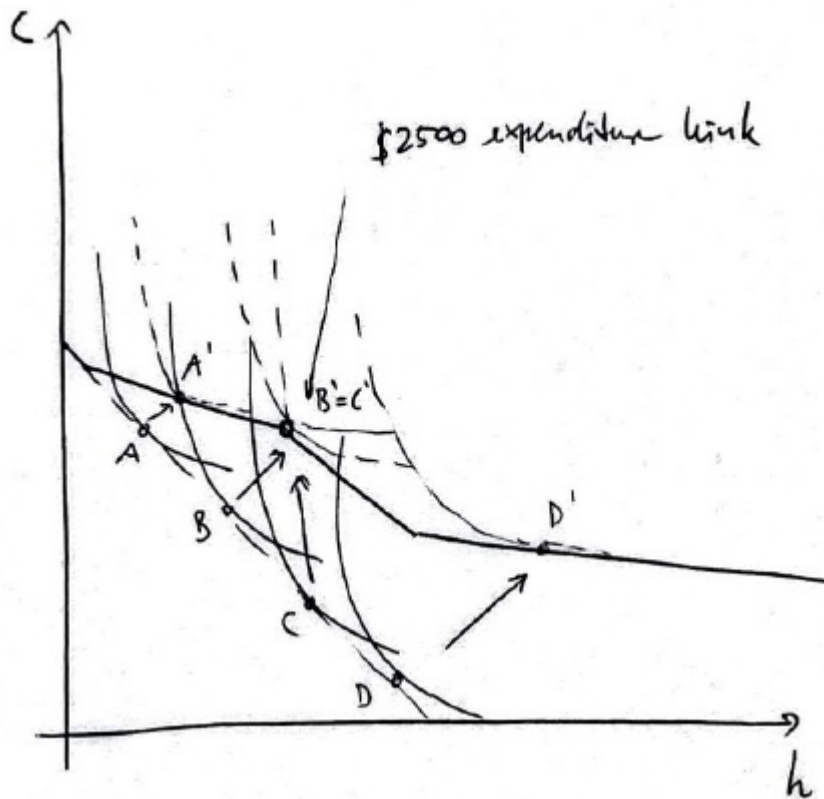
The graph shows healthcare consumption (h) vs. all other goods (c). p is the relative price of healthcare, normalizing the price of c to 1. Compared to a situation with no insurance, an insurance plan with a 50% coinsurance rate leads to an outward rotation of the budget line as shown in the graph. Under the standard assumptions (healthcare as a normal good), healthcare consumption should increase as a result of being insured.

c.

i. Budget constraint with the insurance scheme: The slope is -1 for the first \$250 of expenditure due to the deductible where the patient has to pay 100% of the marginal cost. For the next \$2250 of spending, the patient only has to pay a coinsurance rate of 25%, so the slope is flatter at $-1/4$. For the next \$3600, the patient again has to pay 100% of the cost, after which he has to pay 5% in the flat, rightmost segment of the budget constraint. Note that we normalize p to 1 in the graph for simplicity.

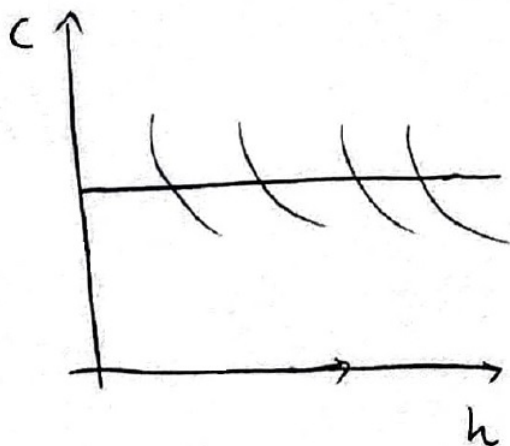


ii. The budget constraint features a convex kink at \$2500 of patient spending, corresponding to $h = 9250$. The graph below illustrates how individuals with different preferences for healthcare respond to the insurance scheme. Individuals A, B, C, D are at different levels of h before the scheme is introduced, reflecting their different preferences. This corresponds to a smooth distribution of healthcare expenditure in the absence of the insurance scheme. With the insurance scheme, the individuals locate at points A', B', C' and D', respectively. Note that both individuals B and C locate exactly at the kink in spite of their different preferences – in fact, individuals with initial h between C and C will all bunch at the kink. Hence, we expect a spike in the distribution of healthcare expenditure at the location of the kink once the insurance scheme is introduced.



d.

It is true that according to a standard model where individuals derive utility from consuming healthcare, healthcare expenditure should become very high or even infinite when the price of healthcare is zero. The graph below illustrates that individuals could reach ever higher levels of utility by increasing healthcare.



But is it really true that individuals would always increase their utility from consuming more healthcare? What aspects of the real world are missing from the simple model? First, there may be some non-monetary costs of seeking healthcare, such that the true private cost

is not zero even if the price is zero. This includes an opportunity cost of time, for instance. Second, it is likely that a point of satiation is reached where $u'(h) \leq 0$ above some level of h .